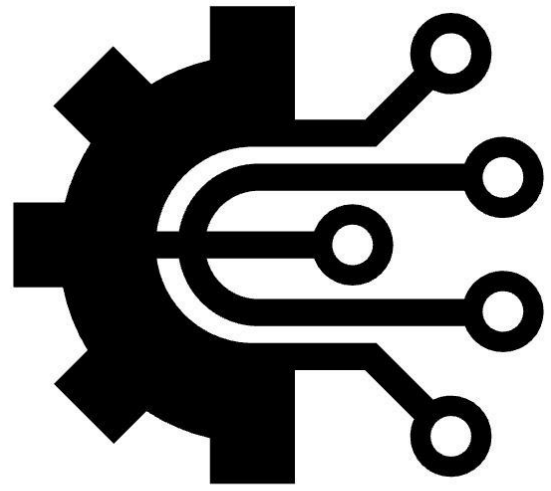




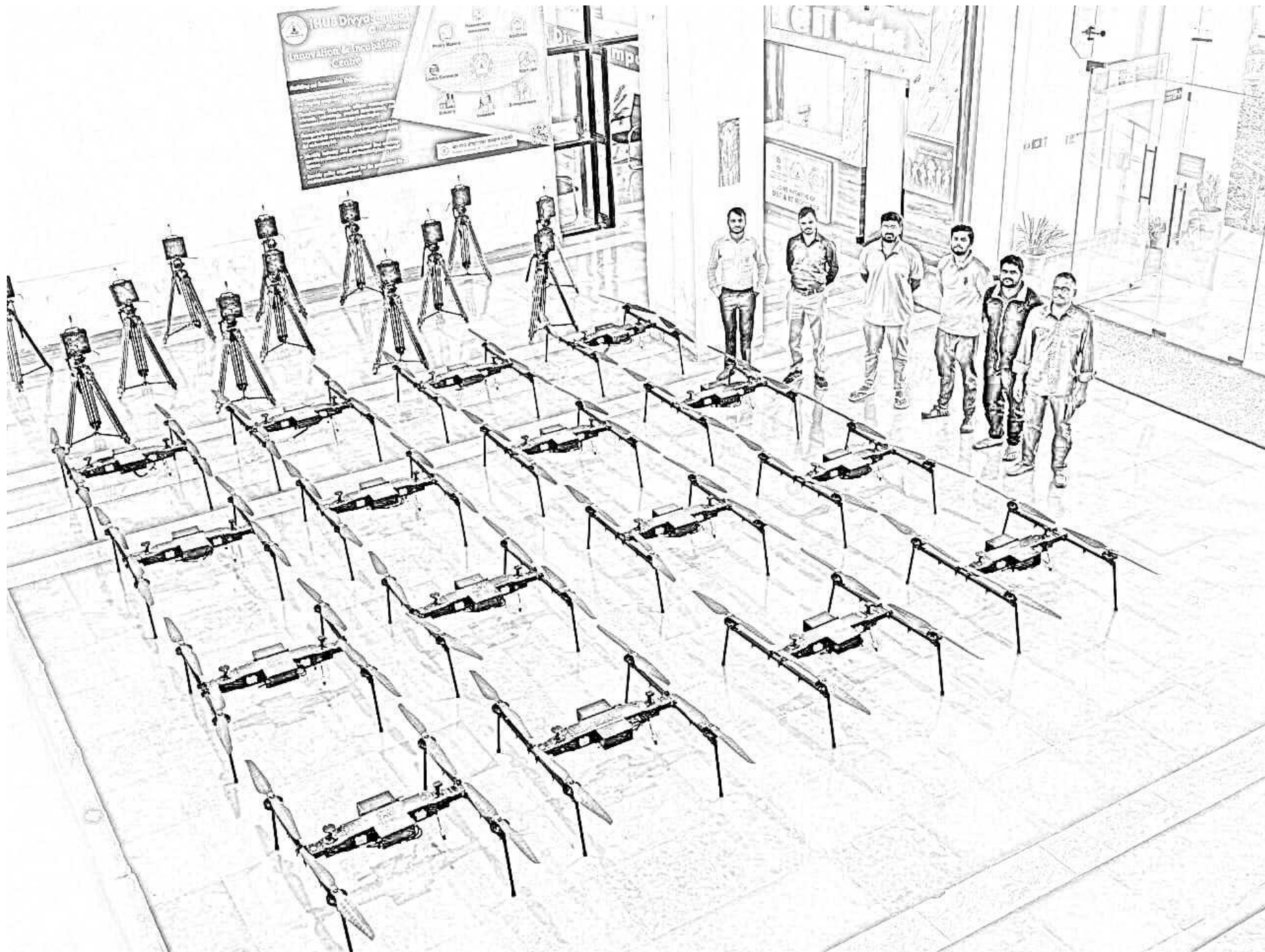
Lenviz Technologies



IIT Roorkee

iDex Open Challenge

By: Prabhat Sharma



Introduction

Lenviz Technologies, a defence-tech startup incubated at IIT Roorkee's Divyasampark i-HUB, is redefining high-altitude military operations by developing advanced autonomous drones, the company provides critical surveillance and GPS denied solutions for extreme environments where traditional methods are risky and insufficient.



GPS Denied - Building Intervention Drone

bidplus.gem.gov.in/bidding/bid/getBidResultView/6771452/participated



(2) - prabha... PDF Compress PDF - q... GitHub - nasa-jpl/... SMPP Peacekeepe... Case Study 03: Dr... Defence Standard...

HOME / RA STATUS

1. RA DETAILS - GEM/2024/R/370045

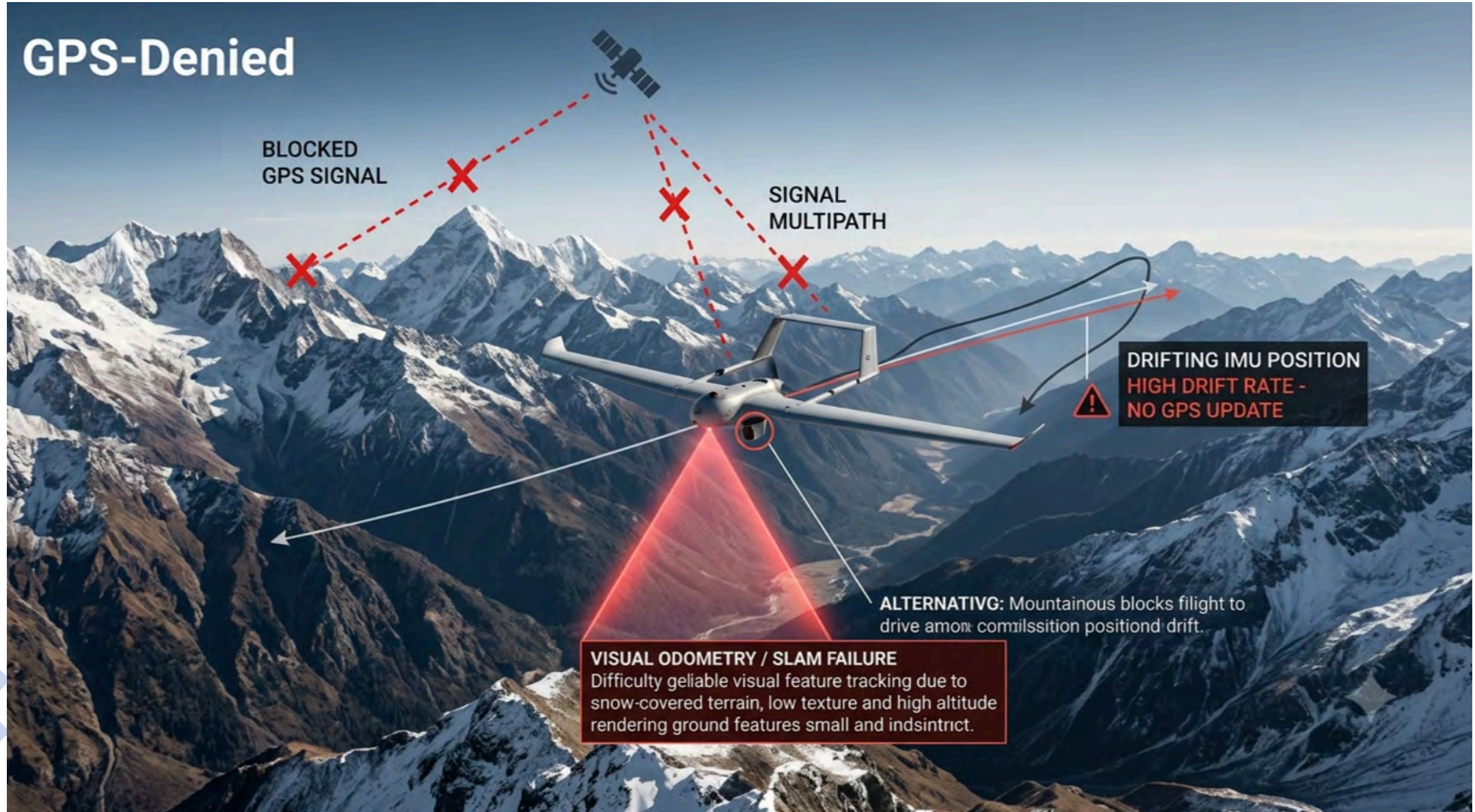
3. FINANCIAL EVALUATION

List Of Sellers Qualified Financially



S.No.	Seller Name	Offered Item	Total Price	Rank
1	DILATON TECHNOLOGIES PRIVATE LIMITED(MSE,MII)	Item Categories : BUILDING INTERVENTION DRONE	₹ [REDACTED]	L1
2	LENVIZ TECHNOLOGIES PRIVATE LIMITED (MSE,MII)	Item Categories : BUILDING INTERVENTION DRONE	₹ [REDACTED]	L2

Problem Statement

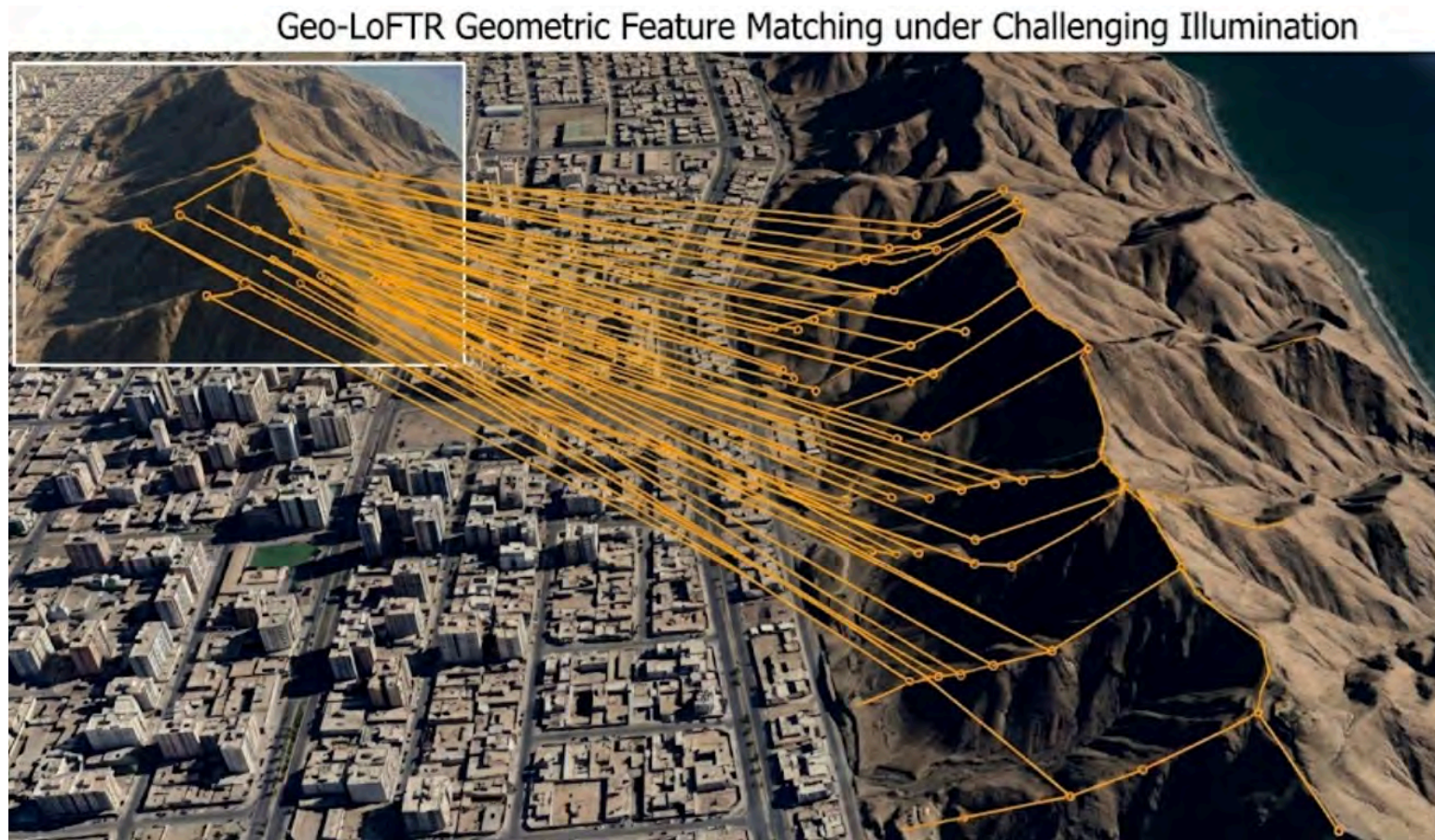


Different Terrain Types



The Geometric Matcher

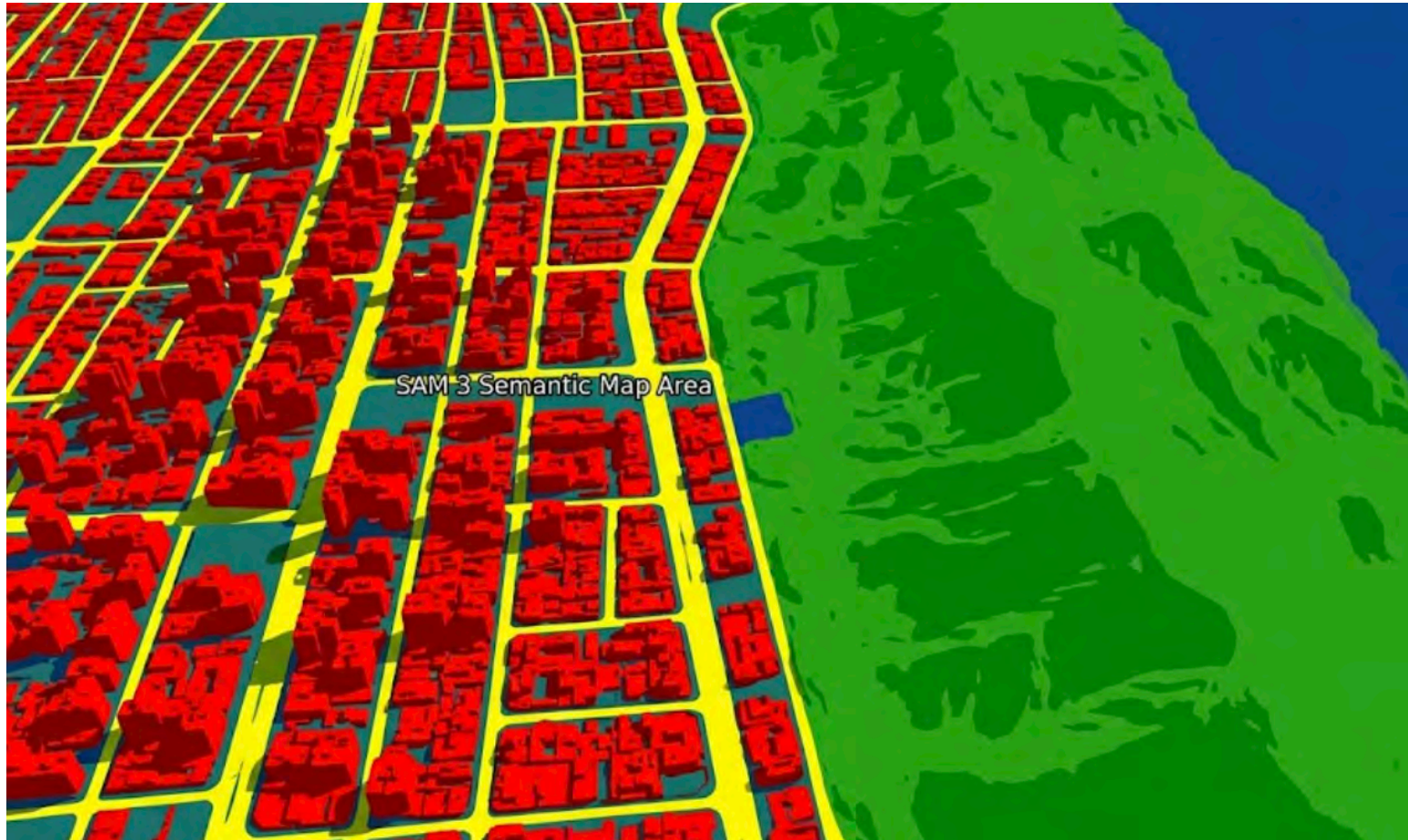
1. **(The "Geometric Matcher"):** This technique directly matches visual features between the drone's camera and a reference map (e.g., satellite imagery or a pre-built 3D model). It's a determined where you are approach. The system's have achieved an average localization error of under 0.1 meters in real-world tests



Cons: Localising to Coordinate level for accuracy

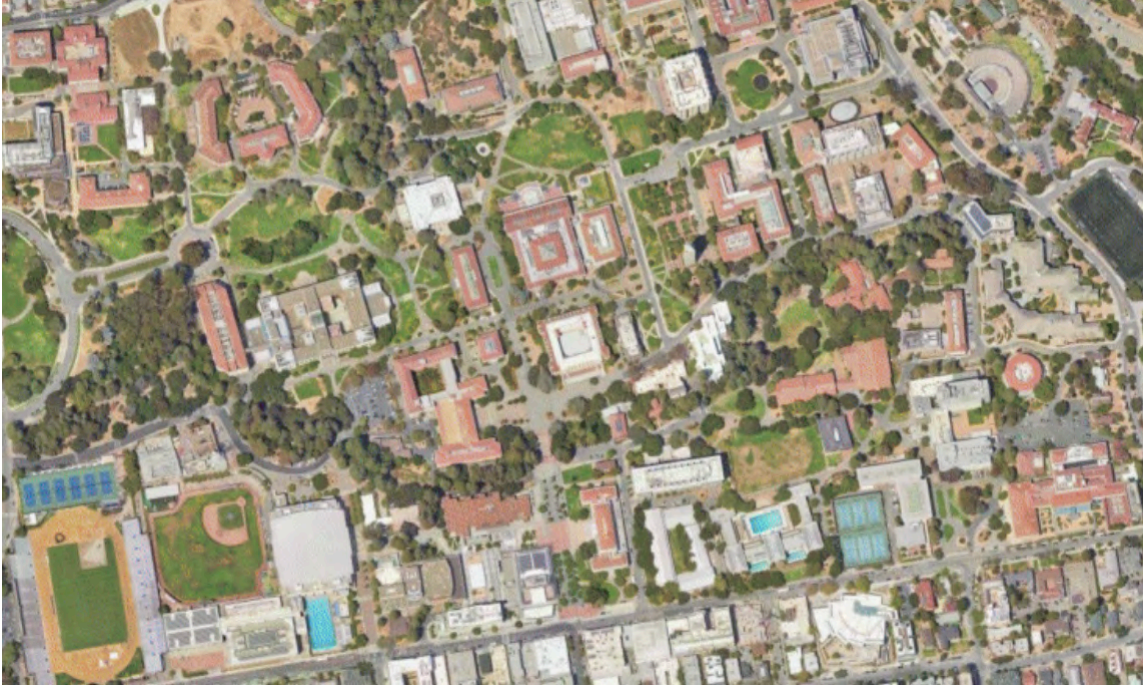
Segmentation Interpreter

The "Semantic Interpreter": SAM3 interprets a scene by identifying and segmenting objects mentioned in a text prompt (e.g., "building" or "roofs"). This creates a semantic map. It's a understand-what-you-see approach, used in research for "visual place recognition" to match a drone's view to a set of known locations



Cons: Each segment is assigned a GPS coordinate

Segmentation Interpreter



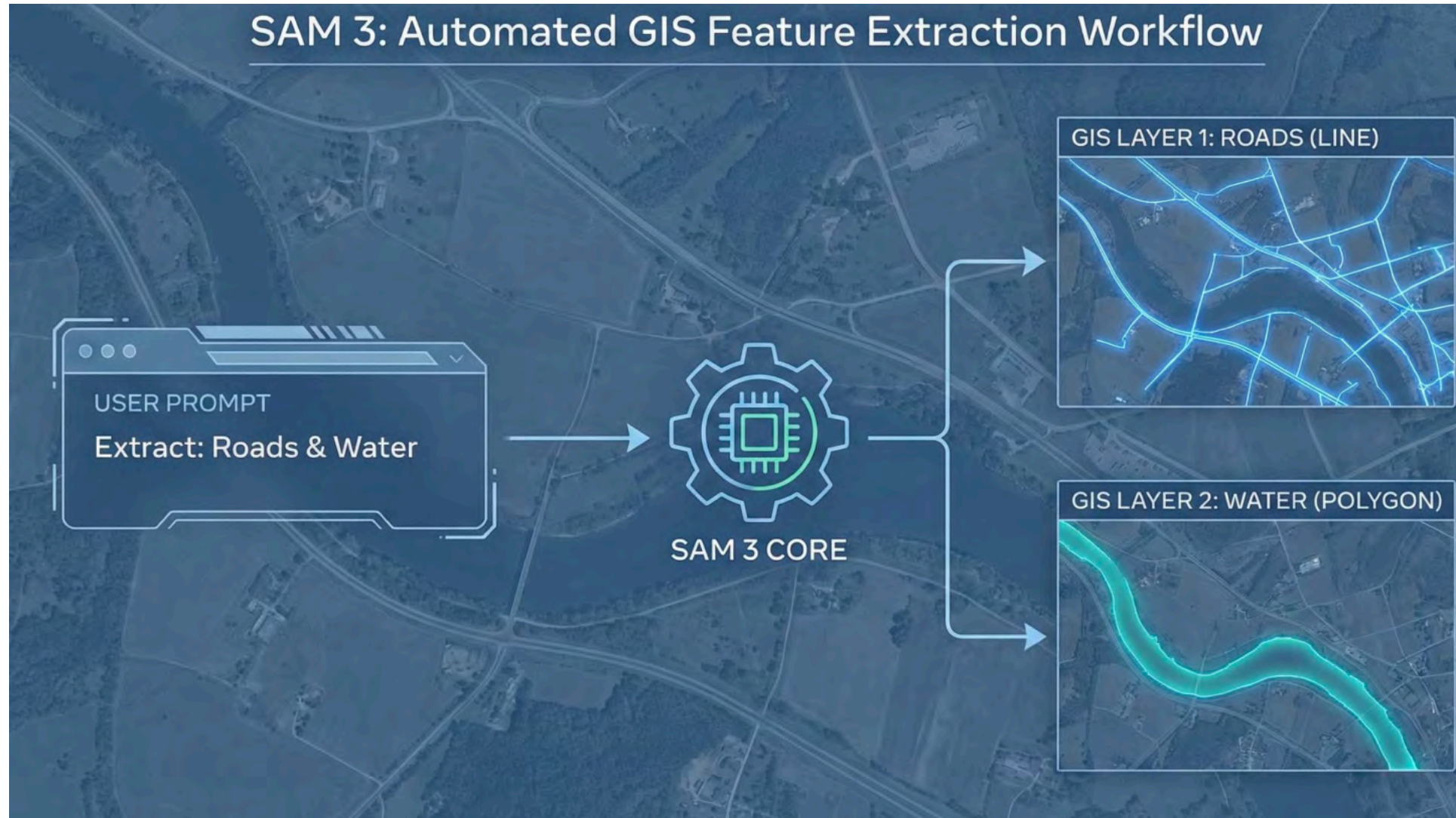
1. Run the segmentation model to generate GPS coordinates of each segment in the Maps used in the Ground Control Station software
2. Create a local database of segments with GPS coordinates
3. Run the segmentation model on live drone imagery and compare with the local database to identify coordinates

Cons: Segmentation works well in structured environments i.e. Urban environment but struggles in unstructured environments such as forests

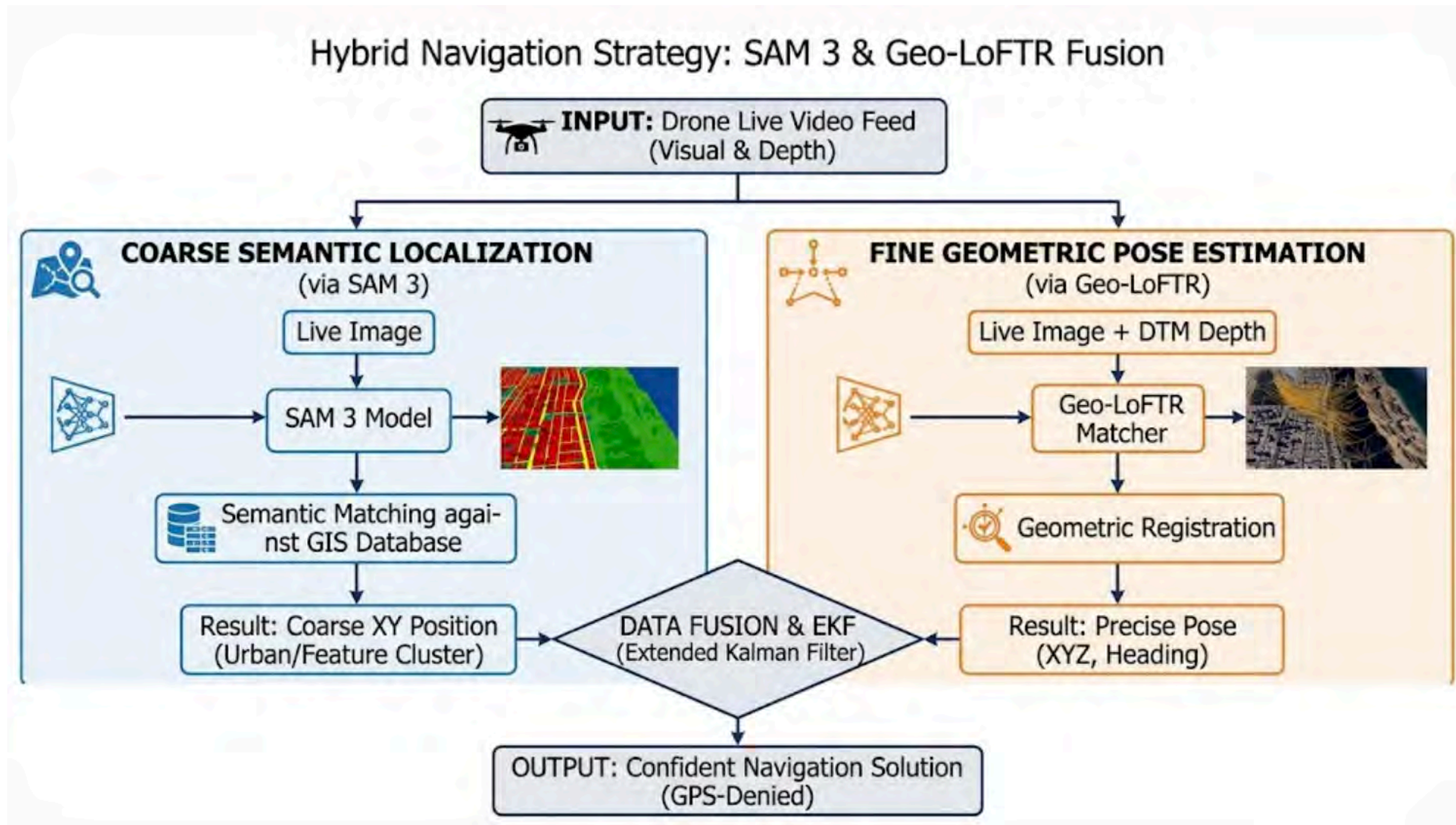
Filtering of useful information - Example



Identifying features in unstructured environments



Fusion of Segmentation & Geo Localization



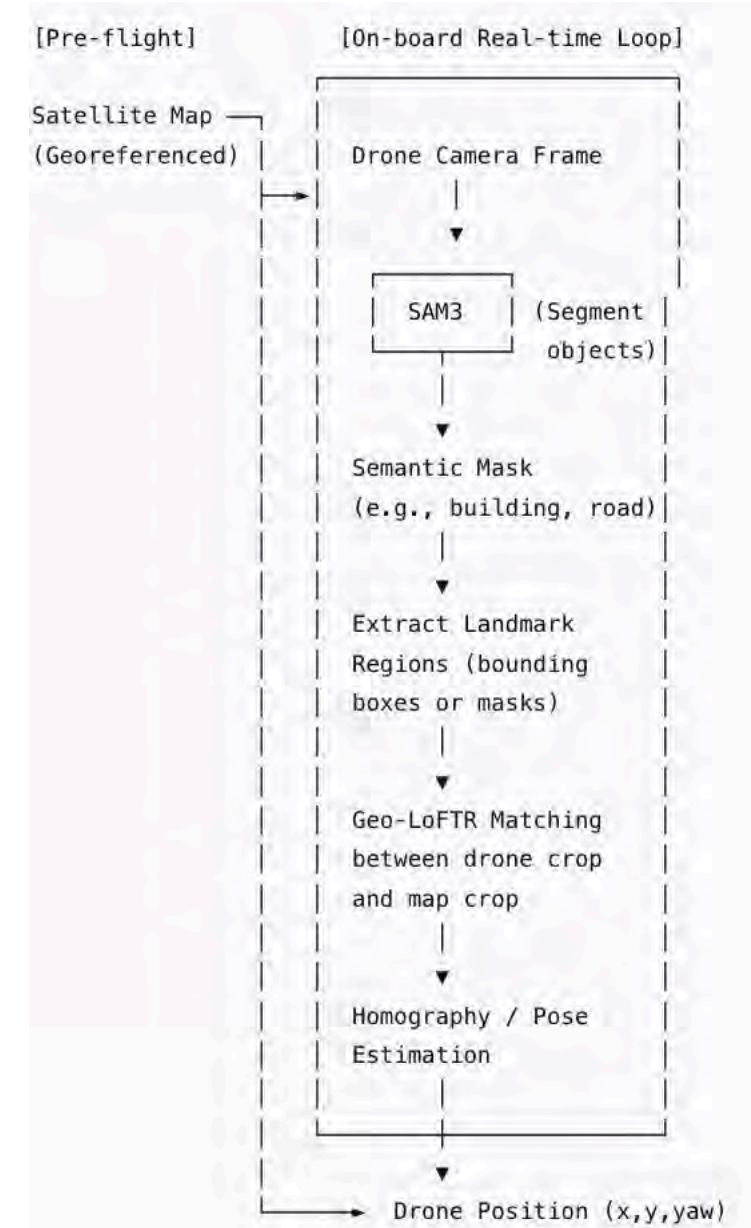
Fusion Benefits

Strategy

By integrating these two systems, the drone gains true multi-modal resilience:

- **Environmental Robustness:** If the drone is flying over a city, SAM 3 (Image 2) provides the anchor, locking onto streets and buildings that are immune to visual texture changes (like snow or new facades).
- **Appearance Robustness:** If the drone is flying over unstructured mountains under a low sun angle (Image 3), Geo-LoFTR provides the anchor, utilising the persistent 3D shape of the ridges rather than their visual texture.

The combination delivers a navigation system that is not only robust against visual drift but also immune to the key failures that typically break Map-based Localisation (appearance change, extreme lighting, and feature scarcity).



Summary of Fusion Roles

Metric / Mechanism	IMU Backbone	SAM 3 Pipeline	Geo-LoFTR Pipeline
Frequency	Extremely High (200 Hz)	Low (1 Hz)	Moderate (10 Hz)
Role in EKF	State Prediction (Fills the gaps between camera frames)	Coarse Measurement Correction (Prevents global bounding drift)	Fine Measurement Correction (Provides local precision & attitude)
Data Provided	Raw acceleration & angular velocity	Absolute 2D/3D global position coordinates	Relative/Absolute 6-DoF position & orientation
Trust Factor (\$R\$)	Trusts implicitly short-term; completely distrusts long-term.	Highly trusted in structured environments; ignored in wilderness.	Highly trusted in rocky/barren geometry; treated carefully in cities.

By coupling the high-speed state propagation of an IMU with the macro-stability of SAM 3 and the micro-precision of Geo-LoFTR, the EKF creates a navigation loop that is virtually impossible to disorient, even in completely dark, changing, or GPS-denied theaters of operation.

Team

Anirudh Putsala

B. Tech Major in Mechanical and Minor in Robotics (IIT - Gandhinagar)

A problem solver and a robotics enthusiast with experience in product designing, electronics, and coding and worked on mobile robots on ground and underwater in which I have built the core algorithms from scratch and implemented them on the fly. My recent works are virtual hand gestures being used to increase accuracy of MANPADs such as IGLA and Zu-23, Kamikaze-strike using a drone, on a selected target in real time.

Akash Patil

M. Tech Embedded system (VIT University, Vellore), B.E Electronics & Communications (Pune University)

Passionate robotics engineer with 8+ years of deep experience of developing low swap embedded platforms for autonomous drones, autonomous mobile robots and robotic arm for the Indian defence sector as well as private sector. Technical lead for building cutting edge indigenous autonomous AI enabled UAV products. Deep experience of working on embedded platform architecture, Autonomous robots navigation, AI/ML, ML based target tracking and ballistic computer serving for private and defence sector.

Prabhat Sharma

MS (BITS Pilani), MCA (PTU), B.Sc. (Delhi University)

Two Decades of IT consulting experience with 6+ years in USA. Technical architect with deep experience/ understanding of Defence domain and has worked with the Indian Military organization in close quarters such as NSG, Army Air Defence etc. Deep experience in working on Autonomous drones, ballistic computer and AI/ML technologies for the Indian defence establishment.



Titan Series of UAVs

We push the boundaries of what is possible with drone technology, specializing in high altitude operations. We are pioneers in developing and deploying drones that soar beyond conventional limitations, unlocking a new era of data collection, inspection, and communication in the most challenging environments.

Note: Winner of NSG Hackathon 2022 for “GPS Denied Unmanned Aerial Vehicle” category

Patent application: “202411061797” HIGH ALTITUDE SURVEILLANCE UAV SYSTEM AND METHOD THEREOF”

Contact: prabhat@lenviz.com



Titan Mini V2

- ❖ High Altitude Surveillance Quadcopter with ceiling limit of **6500m AMSL & 1500m AGL**
- ❖ Multipurpose drone with deliveries supported up-to 5 KG
- ❖ Low aural signature (less than 40db at 300 meters) and high endurance makes it perfect for CI/CT operations
- ❖ Weight - 10 kg
- ❖ Endurance - 90 min at MSL
- ❖ 60 min at 6500 m
- ❖ Max speed - 15 m/s
- ❖ Range - 15 km
- ❖ 128 / 256-bit encryption
- ❖ EO - 1920 * 1080 pix / 20 x optical zoom
- ❖ IR - 640 * 420 pix / 4 x digital zoom
- ❖ **Logistic Delivery of 1 KG payload/munition with 45 min endurance at 6000 meters**
- ❖ **DOOAF capability**

Note: Delivered Titan Mini UAVs to the Northern Command Contract Id - GEMC-511687704429556 in Nov 2024



Titan Mini V3

- ❖ Best in class proven endurance of
 - **90 min at 6500 meters**
 - 120 min at MSL
- ❖ Low aural signature (less than 40db at 300 meters) and high endurance makes it perfect for CI/CT operations
- ❖ Weight - 11 kg
- ❖ Max speed - 15 m/s
- ❖ Range - 25 km
- ❖ 128 / 256-bit encryption
- ❖ EO - 1920 * 1080 pix / 30 x optical zoom (Person detection from 3-5 KM range)
- ❖ IR - 640 * 420 pix / 8 x digital zoom (Person detection 1000 m away)
- ❖ **Logistic Delivery of 1 KG payload/munition at 6500 meters**
- ❖ **DOOAF capability**

Note: Cleared 2 TECs (Northern Command)

L1 for GEM/2024/B/5406170 (RA No. GEM/2025/ R/457357)





Thank You

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